

Monadic second-order logic with cardinality predicates

OPG-37440	L1 - Tractable	Partiellement résolu	Logic
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The problem concerns the extension of Monadic Second Order Logic (over a binary relation representing the edge relation) with the following atomic formulas:

- $\text{Card}(X) = \text{Card}(Y)$
- $x \text{ belongs to } A$

where A is a fixed recursive set of integers.

Let us fix k and a closed formula F in this language.

Conjecture Is it true that the validity of F for a graph G of tree-width at most k can be tested in polynomial time in the size of G ?

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